Experimental versus Theoretical Probability from C.M.P.

20. If you drop a tack on the floor, there are two possible outcomes: the tack lands on its side (point down), or the tack lands on its head (point up). Kalifa dropped a tack 100 times. The table shows the results.



point down point up

Outcomes	Number of Times It Occurs
Tack lands point up	58
Tack lands point down	42

a. Suppose you drop Kalifa's tack 500 times. How many times do you expect it to land point up?

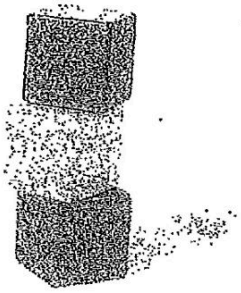
b. Is it equally likely that the tack will land point up or point down? Explain.

c. Is it possible to determine theoretical probabilities for this situation? Why or why not?

22. A bucket contains 60 marbles. Some are red, some are blue, and some are white. The probability of drawing a red marble is 35%. The probability of drawing a blue marble is 25%. How many marbles of each color are in the bucket?

Connections

23. Hannah's teacher brought in a bucket containing 72 blocks. The blocks are red, yellow, or blue. Hannah wants to figure out the number of blue blocks without emptying the bucket.



Hannah chooses a block from the bucket, records its color, and then replaces it. Of her 14 draws, she records blue 5 times. Based on Hannah's experiment, how many of the blocks are blue? Explain.

24. Suppose you roll two number cubes. What is the probability that the product of the numbers will be a multiple of 5?

25. If you roll two number cubes 100 times, about how many times can you expect the product of the numbers to be a multiple of 5?

26. Suppose you roll two number cubes. What is the probability that the product of the numbers is a multiple of 7?

27. If you roll two number cubes a million times, about how many times can you expect the product of the numbers to be a multiple of 7?

28. Suppose you roll two number cubes and multiply the numbers. Find each probability.

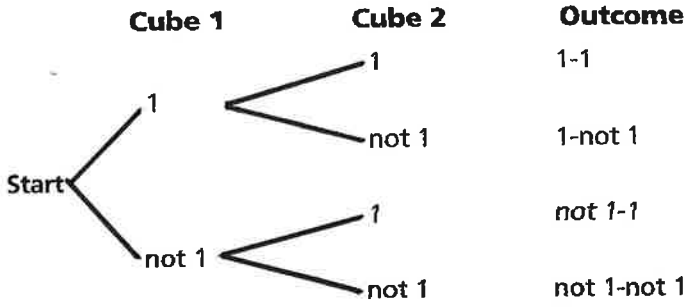
a. *P*(product is a multiple of 3 and 4)

b. *P*(product is a multiple of 3 or 4)

c. P(product has a factor of 5 and 3)

Extensions - Challenge Problems

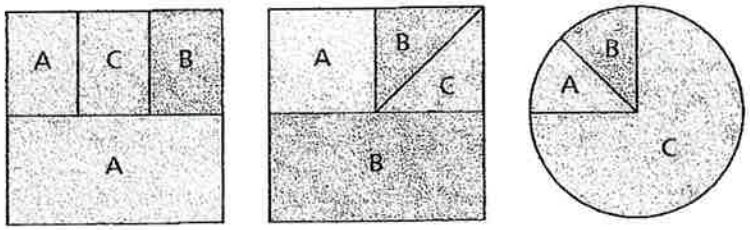
29. Tricia wants to determine the probability of getting two 1's when two number cubes are rolled. She makes a tree diagram and uses it to list the possible outcomes.



She says that, because there are four possible outcomes, the probability of getting 1 on both number cubes is $\frac{1}{4}$. Is Tricia right? Explain.

30. Juan invented a two-person game in which players take turns rolling three number cubes. If the sum is even, Player A gets a point. If the sum is odd, Player B gets a point. Is Juan's game a fair game? Explain.

For Exercises 31–33, a computer places a dot at random on each dartboard below.



Board 1 **Board 2** **Board 3**

31. For each dartboard, what is the probability that a dot will be in a region marked A? A region marked B? A region marked C?

32. For Board 1, what is the probability that a dot will be in a region marked A or B?

33. For Board 2, what is the probability that a dot will *not* be in region C?

Experimental Versus Theoretical Probability from CPM

20. If you drop a tack on the floor, there are two possible outcomes: the tack lands on its side (point down), or the tack lands on its head (point up). Kalifa dropped a tack 100 times. The table shows the results.



point down



point up

Outcomes	Number of Times It Occurs
Tack lands point up	58
Tack lands point down	42

a. Suppose you drop Kalifa's tack 500 times. How many times do you expect it to land point up? $58 \cdot 5 = 290$

b. Is it equally likely that the tack will land point up or point down? Explain. No, not based on the results shown above.

c. Is it possible to determine theoretical probabilities for this situation? Why or why not? No, we only have experimental probability data here.

22. A bucket contains 60 marbles. Some are red, some are blue, and some are white. The probability of drawing a red marble is 35%. The probability of drawing a blue marble is 25%. How many marbles of each color are in the bucket? 21 red, 15 blue, 24 white

23. Hannah's teacher brought in a bucket containing 72 blocks. The blocks are red, yellow, or blue. Hannah wants to figure out the number of blue blocks without emptying the bucket.



Hannah chooses a block from the bucket, records its color, and then replaces it. Of her 14 draws, she records blue 5 times. Based on Hannah's experiment, how many of the blocks are blue? Explain. $\frac{5}{14} = \frac{?}{72}$ → about 26

24. Suppose you roll two number cubes. What is the probability that the product of the numbers will be a multiple of 5? $\frac{11}{36}$ (Use a multiplication table.)

25. If you roll two number cubes 100 times, about how many times can you expect the product of the numbers to be a multiple of 5? $\frac{11}{36} = \frac{?}{100}$ → about 31

26. Suppose you roll two number cubes. What is the probability that the product of the numbers is a multiple of 7? $\frac{0}{36}$ or 0%

27. If you roll two number cubes a million times, about how many times can you expect the product of the numbers to be a multiple of 7? 0 or 0%

28. Suppose you roll two number cubes and multiply the numbers. Find each probability.

a. P(product is a multiple of 3 and 4) $\frac{7}{36}$

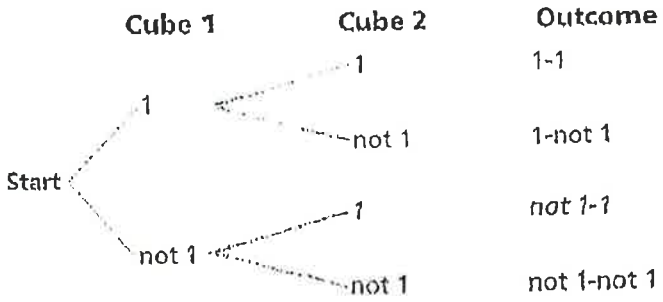
b. P(product is a multiple of 3 or 4) → $\frac{28}{36} = \frac{7}{9}$

c. P(product has a factor of 5 and 2)

Connections

Extensions - Challenge Problems

29. Tricia wants to determine the probability of getting two 1's when two number cubes are rolled. She makes a tree diagram and uses it to list the possible outcomes.



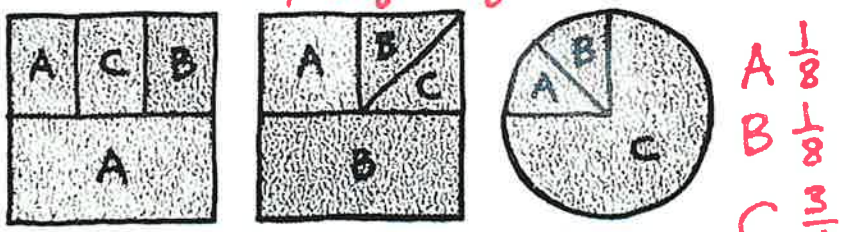
She says that, because there are four possible outcomes, the probability of getting 1 on both number cubes is $\frac{1}{4}$. Is Tricia right? Explain. No, she needs 5 "not 1" branches because there are 5 numbers on a die that are not 1.

30. Juan invented a two-person game in which players take turns rolling three number cubes. If the sum is even, Player A gets a point. If the sum is odd, Player B gets a point. Is Juan's game a fair game? Explain. Yes, the sum will be odd half the time and even half the time.

For Exercises 31–33, a computer places a dot at random on each dartboard below.

A $\frac{1}{4}$ B $\frac{5}{8}$ C $\frac{1}{8}$

A $\frac{2}{3}$
B $\frac{1}{6}$
C $\frac{1}{6}$



Board 1 Board 2 Board 3

A $\frac{1}{8}$
B $\frac{1}{8}$
C $\frac{3}{4}$

31. For each dartboard, what is the probability that a dot will be in a region marked A? A region marked B? A region marked C?

32. For Board 1, what is the probability that a dot will be in a region marked A or B? $\frac{5}{6}$

33. For Board 2, what is the probability that a dot will *not* be in region C? $\frac{7}{8}$

46

Name: _____ Date _____

Topic : Probability Word Problems- Worksheet 1

What is the probability?

1. Jill is playing cards with her friend when she draws a card from a pack of 20 cards numbered from 1 to 20. What is the probability of drawing a number that is square?

2. Each of the letters in the word SAMSUNG are on separate cards, face down on the table. If you pick a card at random, what is the probability that its letter will be S or U?

3. A magician showed a magic trick where he picked one card from a standard deck. Determine what the probability is that the card will be a queen card?

4. A bag contains ten black marbles, twenty white marbles, and five grey marbles. You pick one without looking. What is the probability that the marble will be either white OR black?

5. You ask a friend to think of a number from four to twelve. What is the probability that his number will be 8?

6. Each of letters in the word are on separate cards, face down on the table. If you pick a card at random, what is the probability that its letter will be L or E?

7. You roll a SIX sided die. What is the probability that the value of the roll will be one?

8. A bag contains 5 blue sticks, 4 red sticks, and 3 orange sticks and you ask a friend to pick one without looking. What is the probability that the stick will be blue?

9. You think of a number from the first twenty negative integers. What is the probability that the integer chosen will be divisible by 4?

10. When a six sided die is rolled then what is the probability that the number rolled will be five?



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Kuta Software - Infinite Algebra 2 Name_____

Independent and Dependent Events

Date_____ Period____

Determine whether the scenario involves independent or dependent events.

1) You flip a coin and then roll a fair six-sided die. The coin lands heads-up and the die shows a one.

2) A bag contains eight red marbles and four blue marbles. You randomly pick a marble and then pick a second marble without returning the marbles to the bag. The first marble is red and the second marble is blue.

3) A box of chocolates contains five milk chocolates, five dark chocolates, and five white chocolates. You randomly select and eat three chocolates. The first piece is milk chocolate, the second is dark chocolate, and the third is white chocolate.

4) A cooler contains ten bottles of sports drink: four lemon-lime flavored, three orange flavored, and three fruit-punch flavored. Three times, you randomly grab a bottle, return the bottle to the cooler, and then mix up the bottles. The first time, you get a lemon-lime drink. The second and third times, you get fruit-punch.

Find the probability.

5) You flip a coin and then roll a fair six-sided die. The coin lands heads-up and the die shows an even number.

6) You roll a fair six-sided die twice. The first roll shows a five and the second roll shows a six.

7) There are eight shirts in your closet, four blue and four green. You randomly select one to wear on Monday and then a different one on Tuesday. You wear blue shirts both days.

8) A basket contains five apples and seven peaches. You randomly select one piece of fruit and eat it. Then you randomly select another piece of fruit. The first piece of fruit is an apple and the second piece is a peach.

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Name: KEY Date _____

Topic : Probability Word Problems- Worksheet 1

What is the probability?

$1^2=1,\ 2^2=4,\ 3^2=9,\ 4^2=16$

1. Jill is playing cards with her friend when she draws a card from a pack of 20 cards numbered from 1 to 20. What is the probability of drawing a number that is square?
↳ 1, 4, 9, 16 $\frac{4}{20} = \frac{1}{5}$

2. Each of the letters in the word SAMSUNG are on separate cards, face down on the table. If you pick a card at random, what is the probability that its letter will be S or U? $\frac{3}{7}$

3. A magician showed a magic trick where he picked one card from a standard deck. Determine what the probability is that the card will be a queen card? $\frac{4}{52} = \frac{1}{13}$

4. A bag contains ten black marbles, twenty white marbles, and five grey marbles. You pick one without looking. What is the probability that the marble will be either white OR black? $\frac{30}{35} = \frac{6}{7}$

5. You ask a friend to think of a number from four to twelve. What is the probability that his number will be 8? $\frac{1}{9}$

6. Each of letters in the word are on separate cards, face down on the table. If you pick a card at random, what is the probability that its letter will be L or E?

7. You roll a SIX sided die. What is the probability that the value of the roll will be one? $\frac{1}{6}$

8. A bag contains 5 blue sticks, 4 red sticks, and 3 orange sticks and you ask a friend to pick one without looking. What is the probability that the stick will be blue? $\frac{5}{12}$

9. You think of a number from the first twenty negative integers. What is the probability that the integer chosen will be divisible by 4? $-4, -8, -12, -16, -20$ $\frac{5}{20} = \frac{1}{4}$

10. When a six sided die is rolled then what is the probability that the number rolled will be five? $\frac{1}{5}$

Kuta Software - Infinite Algebra 2

Name KEY

Independent and Dependent Events

Date _____ Period ____

Determine whether the scenario involves independent or dependent events.

1) You flip a coin and then roll a fair six-sided die. The coin lands heads-up and the die shows a one.

independent

2) A bag contains eight red marbles and four blue marbles. You randomly pick a marble and then pick a second marble without returning the marbles to the bag. The first marble is red and the second marble is blue.

dependent

3) A box of chocolates contains five milk chocolates, five dark chocolates, and five white chocolates. You randomly select and eat three chocolates. The first piece is milk chocolate, the second is dark chocolate, and the third is white chocolate.

dependent

4) A cooler contains ten bottles of sports drink: four lemon-lime flavored, three orange flavored, and three fruit-punch flavored. Three times, you randomly grab a bottle, return the bottle to the cooler, and then mix up the bottles. The first time, you get a lemon-lime drink. The second and third times, you get fruit-punch.

independent

Find the probability.

5) You flip a coin and then roll a fair six-sided die. The coin lands heads-up and the die shows an even number.

$\frac{1}{2} \cdot \frac{3}{6} = \frac{3}{12} = \boxed{\frac{1}{4}}$ (or $\frac{1}{2}$)

6) You roll a fair six-sided die twice. The first roll shows a five and the second roll shows a six.

$\frac{1}{6} \cdot \frac{1}{6} = \boxed{\frac{1}{36}}$

7) There are eight shirts in your closet, four blue and four green. You randomly select one to wear on Monday and then a different one on Tuesday. You wear blue shirts both days.

$\frac{4}{8} \cdot \frac{3}{7} = \frac{12}{56} = \boxed{\frac{3}{14}}$ (or $\frac{1}{2}$)

8) A basket contains five apples and seven peaches. You randomly select one piece of fruit and eat it. Then you randomly select another piece of fruit. The first piece of fruit is an apple and the second piece is a peach.

$\frac{5}{12} \cdot \frac{7}{11} = \boxed{\frac{35}{132}}$

7th grade math

you are: _____
date: _____

Fraction Operations

Please show your work on each problem.

1. Simplify these fractions.

a. $\frac{6}{21}$

b. $\frac{24}{40}$

2. Multiply these fractions. Show your work and simplify your answer for full credit.

a. $\frac{2}{3} \cdot \frac{1}{8}$

b. $2\frac{5}{6} \cdot 3\frac{1}{3}$

3. Divide these fractions! Your final answer should be fully simplified.

a. $\frac{2}{3} \div \frac{1}{8}$

b. $2\frac{5}{6} \div 3\frac{1}{3}$

7th grade math

you are: _____
date: _____

4. Add or subtract. Your final answer should be fully simplified.

a. $\frac{1}{2}+\frac{5}{7}$

b. $\frac{4}{5}-\frac{1}{6}$

c. $1\frac{1}{5}+\frac{3}{8}$

d. $\frac{13}{2}+\frac{3}{3}$

5. Fractions, Decimals & Percents

a. Convert 5/8 to a decimal.

b. Convert 0.03 to a percentage.

c. Convert 85% to a reduced fraction.

7th grade math

you are: _____
date: _____

Fraction Operations

Please show your work on each problem.

1. Simplify these fractions.

a. $\frac{6}{21}$ $\frac{2}{7}$

b. $\frac{24}{40}$ $\frac{3}{5}$

2. Multiply these fractions. Show your work and simplify your answer for full credit.

a. $\frac{2}{3} \cdot \frac{1}{8}$ $\frac{1}{4} = \frac{1}{12}$

b. $2\frac{5}{6} \cdot 3\frac{1}{3}$

$\frac{17}{6} \cdot \frac{10}{3} = \frac{170}{18} = \frac{85}{9}$

3. Divide these fractions! Your final answer should be fully simplified.

a. $\frac{2}{3} \div \frac{1}{8}$

$\frac{2}{3} \cdot \frac{8}{1} = \frac{16}{3}$

b. $2\frac{5}{6} \div 3\frac{1}{3}$

$\frac{17}{6} \div \frac{10}{3}$

$\frac{17}{6} \cdot \frac{3}{10} = \frac{51}{60} = \frac{17}{20}$

14

7th grade math

you are: _____
date: _____

4. Add or subtract. Your final answer should be fully simplified.

a. $\frac{1}{2}+\frac{5}{7}$

$\frac{7}{14}+\frac{10}{14}=\boxed{\frac{17}{14}}$

b. $\frac{4}{5}-\frac{1}{6}$

$\frac{24}{30}-\frac{5}{30}=\boxed{\frac{19}{30}}$

c. $1\frac{1}{5}+\frac{3}{8}$

$\frac{6}{5}+\frac{3}{8}$

$\frac{48}{40}+\frac{15}{40}=\boxed{\frac{63}{40}}$

d. $\frac{13}{2}+\frac{3}{3}$

$\frac{39}{6}+\frac{6}{6}=\frac{45}{6}=\boxed{\frac{15}{2}}$

5. Fractions, Decimals & Percents

a. Convert 5/8 to a decimal. 0.625

b. Convert 0.03 to a percentage. 3%

c. Convert 85% to a reduced fraction. $\frac{85}{100}=\boxed{\frac{17}{20}}$

Name _____ Date _____

Probability with Compound Events (Independent and Dependent)
Practice

Describe the events by writing **I** for *independent event* or **D** for *dependent event.*

1. Ann draws a colored toothpick from a jar. Without replacing it, she draws a second toothpick. _____
2. John rolls a six on a number cube and then flips a coin that comes up heads. _____
3. Susie draws a card from a deck of cards and replaces it. She then draws a second card. _____
4. Seth draws a colored tile from a bag, replaces it; draws a second tile from the bag, replaces it; and then draws a tile a third time from the bag. _____
5. You draw a red marble from a bag, and then another red marble (without replacing the first marble)? ____

Using the two spinners, find each **compound** probability.





6. P(A and 2) _____ 7. P(D and 1) _____ 8. P(B and 3) _____

9. P(A and not 2) _____

A box contains 3 red marbles, 6 blue marbles, and 1 white marble. The marbles are selected at random, one at a time, and are **not replaced.** Find each **compound** probability.

10. P(blue and red) _____ 11. P(blue and blue) _____ 12. P(red and white and blue) _____

13. P(red and red and red) _____ 14. P(white and red and white) _____

Suppose that two tiles are drawn from the collection shown at the right. The first tile is replaced before the second is drawn. Find each **compound** probability.

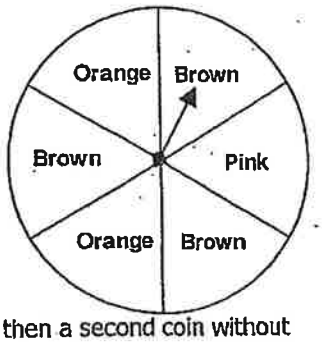
A	R	R	R	C
A	R	R	R	C
E	E	E	E	C

15. P(A and A) _____ 16. P(R and C) _____ 17. P(A and not R) _____

Suppose that two tiles are drawn from the same collection shown above. The first tile is **not** replaced before the second is drawn. Find each **compound** probability.

18. P(A and A) _____ 19. P(R and C) _____ 20. P(A and not R) _____

Use the spinner to the right for the next two problems.



21. If you spin the spinner twice, what is the probability of spinning orange then brown? _____

22. If you spin the spinner twice, what is the probability of spinning brown both times? _____

23. Kevin had 6 nickels and 4 dimes in his pocket. If he took out one coin and then a second coin without replacing the first coin ---
    (a) what is the probability that both coins were nickels? _____
    (b) what is the probability that both coins were dimes? _____
    (b) what is the probability that the first coin was a nickel and the second a dime? _____

Be Sure to Show ALL Your Work!!!

Special Note: Reduce all fractions to lowest terms.

1. A card is chosen at random from a deck of 52 cards. It is then *replaced* and a second card is chosen. What is the probability of choosing a Jack first and a 3 second?

2. What is the probability that from a normal 52 card deck, you randomly draw a 3, and then *without replacing* the 3, you draw the Queen of Hearts?

3. A jar contains 6 red balls, 3 green balls, 5 white balls and 7 yellow balls. Two balls are chosen from the jar, with replacement. What is the probability that both balls chosen are green?

4. A box contains a penny, a nickel, and a dime. Find the probability of choosing a dime first and then, without replacing the dime, choosing a penny.

5. A coin is tossed and a single 6-sided die is rolled. Find the probability of landing on the head side of the coin and rolling 3 on the die.

6. In a class, 12 students are wearing pants and 16 students are wearing shorts. The teacher of the class needs to pick two volunteers. She randomly selects one student, and then selects another student from the class. Find the probability that:

   a. she chooses a student wearing shorts first, then a student wearing pants

   b. she chooses two students wearing pants

2021 KEY

Probability With Compound Events

1. D
2. I
3. I
4. I
5. D
6. $\frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$
7. $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$
8. $\frac{1}{4} \cdot \frac{1}{6} = \frac{1}{24}$
9. $\frac{1}{\cancel{4}_2} \cdot \frac{\cancel{2}^1}{3} = \frac{1}{6}$
10. $\frac{\cancel{6}^3}{\cancel{10}_5} \cdot \frac{\cancel{3}^1}{\cancel{9}_3} = \frac{1}{5}$
11. $\frac{\cancel{6}^2}{\cancel{10}_2} \cdot \frac{\cancel{5}^1}{\cancel{9}_3} = \frac{1}{3}$
12. $\frac{3}{10} \cdot \frac{1}{9} \cdot \frac{6}{8} = \frac{1}{40}$
13. $\frac{3}{10} \cdot \frac{2}{9} \cdot \frac{1}{8} = \frac{1}{120}$
14. $\frac{1}{10} \cdot \frac{3}{9} \cdot \frac{0}{8} = 0$
15. $\frac{2}{15} \cdot \frac{2}{15} = \frac{4}{225}$
16. $\frac{6}{15} \cdot \frac{3}{15} = \frac{2}{25}$
17. $\frac{2}{15} \cdot \frac{9}{15} = \frac{2}{25}$
18. $\frac{2}{15} \cdot \frac{1}{14} = \frac{1}{105}$
19. $\frac{\cancel{6}^3}{\cancel{15}_5} \cdot \frac{\cancel{3}^1}{\cancel{14}_7} = \frac{\cancel{3}}{\cancel{70}} \frac{3}{35}$
20. $\frac{2}{15} \cdot \frac{\cancel{8}^4}{\cancel{14}_7} = \frac{\cancel{3}}{\cancel{35}} \frac{8}{105}$
21. $\frac{2}{6} \cdot \frac{3}{6} = \frac{1}{6}$
22. $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
23. a. $\frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$
    b. $\frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15}$
    c. $\frac{6}{10} \cdot \frac{4}{9} = \frac{4}{15}$

KEY

Be Sure to Show ALL Your Work!!!

Special Note: Reduce all fractions to lowest terms.

1. A card is chosen at random from a deck of 52 cards. It is then *replaced* and a second card is chosen. What is the probability of choosing a Jack first and a 3 second?

$\left(\frac{4}{52}\right)\left(\frac{4}{52}\right) = \left(\frac{1}{13}\right)\left(\frac{1}{13}\right) = \boxed{\frac{1}{169}}$ or 0.59%

2. What is the probability that from a normal 52 card deck, you randomly draw a 3, and then *without replacing* the 3, you draw the Queen of Hearts?

$\left(\frac{4}{52}\right)\left(\frac{1}{51}\right) = \left(\frac{1}{13}\right)\left(\frac{1}{51}\right) = \boxed{\frac{1}{663}}$ or 0.15%

3. A jar contains 6 red balls, 3 green balls, 5 white balls and 7 yellow balls. Two balls are chosen from the jar, with replacement. What is the probability that both balls chosen are green?

$\left(\frac{3}{21}\right)\left(\frac{3}{21}\right) = \left(\frac{1}{7}\right)\left(\frac{1}{7}\right) = \boxed{\frac{1}{49}}$ or 2.04%

4. A box contains a penny, a nickel, and a dime. Find the probability of choosing a dime first and then, without replacing the dime, choosing a penny.

$\left(\frac{1}{3}\right)\left(\frac{1}{2}\right) = \boxed{\frac{1}{6}}$ or 16.67%

5. A coin is tossed and a single 6-sided die is rolled. Find the probability of landing on the head side of the coin and rolling 3 on the die.

$\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) = \boxed{\frac{1}{12}}$ or 8.33%

6. In a class, 12 students are wearing pants and 16 students are wearing shorts. The teacher of the class needs to pick two volunteers. She randomly selects one student, and then selects another student from the class. Find the probability that:

   a. she chooses a student wearing shorts first, then a student wearing pants

$\left(\frac{16}{28}\right)\left(\frac{12}{27}\right) = \left(\frac{4}{7}\right)\left(\frac{4}{9}\right) = \boxed{\frac{16}{63}}$ or 25.40%

   b. she chooses two students wearing pants

$\left(\frac{12}{29}\right)\left(\frac{11}{27}\right) = \left(\frac{3}{7}\right)\left(\frac{11}{27}\right) = \boxed{\frac{33}{189}}$ or 17.46% $\rightarrow \boxed{\frac{11}{63}}$ reduced